

DBMS — Lecture 4

Data Abstraction & Data Independence

Learning Objectives

Explain data abstraction.

Understand the three levels of abstraction.

Differentiate physical and logical data independence.

Data Abstraction

Data abstraction hides unnecessary implementation details and shows users only the information they need.

Three Levels of Abstraction

Physical Level: Describes how data is physically stored, including files, blocks, indexes, and storage details.

Logical Level: Describes what data is stored and relationships among data, e.g., STUDENT(RollNo, Name, Branch).

View Level: Describes the portion of the database visible to a particular user.

Three-Level View

View Level → Logical Level → Physical Level

Data Independence

Data independence is the ability to change a lower-level schema without requiring changes at the next higher level.

Types

Physical Data Independence: Changes to physical storage should not affect the logical schema. Example: changing file organization or adding an index.

Logical Data Independence: Changes to the logical schema should not affect external views or application programs.

Comparison

Type	Meaning
Physical Data Independence	Physical-level changes do not affect logical schema.
Logical Data Independence	Logical-level changes do not unnecessarily affect external views.

Key Takeaway

Abstraction hides complexity; data independence allows changes at one level without unnecessarily affecting higher levels.